Comparative analysis of the performance of NASA Landsat OLI/TIRS, ESA Sentinel 2 MSI, and Roskosmos Kanopé V satellites in Klaipeda (Lithuania)

Master 1 GAED – Geomatic and Spatial Modelling

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First term exam

Remote sensing

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I - **Introduction**

Klaipėda ranks as the third largest city in Lithuania. Geographically situated in the western coastal region, the area is generally flat but includes distinctive Lithuanian landscapes: the Curonian Spit with its high sand dunes, and a 90 km coastline featuring sandy beaches and intermittent cliffs. Strategically positioned on the Baltic Sea, Klaipėda serves as a crucial node for domestic and international trade, driven by a prosperous maritime industry and port-related services. It is the site of Lithuania's largest port. The Klaipėda

Functional Urban Area extends over 1,326.1 km² and recorded a population of 246,301 at the end of 2024.

Klaipeda is the oldest city in the country. It was founded in 1252 under the name of Memel through the establishment of Memelburg (Memel Castle). After many wars and battles, the territory was occupied by the Swedes and Russians, and then became one of the refuges of the court and the royal family of Prussia. In 1871, Memel was integrated into the German Empire. However, in 1919, after the 1st World War, the Treaty of Versailles stipulated the creation of an autonomous territory, despite the opinion expressed by the population, a large majority of whom considered themselves "German". This led to a revolt in 1923, allowing the territory to be attached to the Republic of Lithuania.

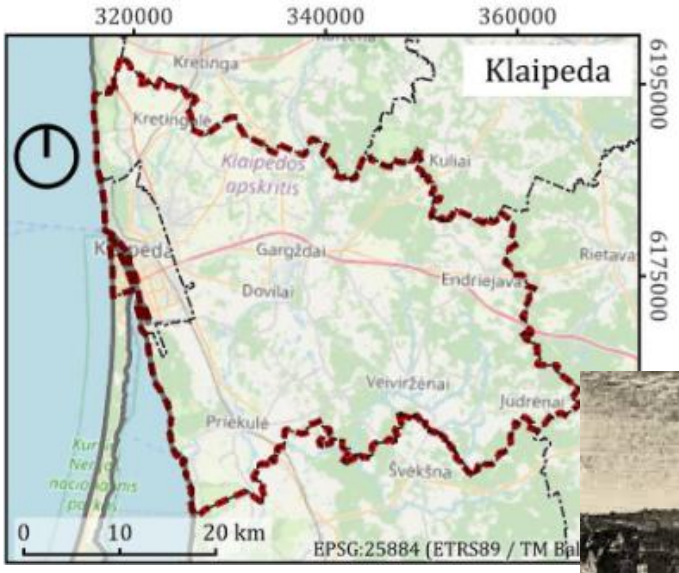


Figure 2 : Area of study : Klaipeda



Figure 1 : Memelburg



Figure 3 : The demolition of the fire department HQ in 1981 was the last of the historical building demolitions in Klaipėda. On the left a Soviet propaganda poster declares: Our work is for thee, oh Fatherland. ©Albinas Stubra.

After World War II, when the territory was ceded to Nazi Germany, Klaipeda was part of the Lithuanian SSR of the USSR. Due to its strategic geography on the coastal zone, the region has experienced strong militarization as well as significant socio-economic changes through agriculture and industry.

The fall of the USSR in 1991 is considered a shock in which the country, integrated into the European Union in 2004, experienced a demographic decline, accompanied by a transition to a market economy in the face of unemployment reaching record figures.



Figure 4 : Curonian Spit

Today, the city is urbanized and metropolised thanks to its major asset that is its port, accompanied by the global process of coastal development, but it can also count on tourism, given its unique history. This duality, between a globalised port economy, the driving force behind coastal development and urban concentration, and the enhancement of a historical and natural heritage for tourism development, draws a territory with complex and often contradictory spatial dynamics. On the one hand, the port generates land pressure, intense logistics flows and artificialization of the coastline; on the other hand, tourism and the quality of the living environment require the preservation of landscapes, coastal ecosystems and built heritage. This constitutive tension places development actors in front of a major challenge: reconciling economic competitiveness, residential and tourist attractiveness, and environmental resilience in a limited and coveted coastal area. To meet this challenge, a fine, up-to-date and spatialized understanding of the processes at work is essential. It is precisely here that remote sensing and spatial analysis come into their own. Satellite images, through their ability to provide synoptic, multi-temporal and multispectral data, offer a privileged observatory to decipher the spatial signatures of these dynamics. They make it possible to quantify urban sprawl, to monitor the evolution of land-sea interfaces, to measure port activity, or to assess the state of vegetation and recreational areas.

Thus, the transition from the description of the issues to their operationalization for public decision-making requires a methodological leap. This leads us to formulate the central problem of this work.

How can a combination of satellite vectors generate concrete geographical information and territorial knowledge essential for the planning and development challenges of Klaipéda?

To answer this question, we will first carry out a detection, recognition and identification of urban objects and structures, then in a second part, we will make a temporal analysis of recent changes in Klaipėda

II - State of the art

To carry out this work, we used 2 sets of data which are satellite images and Open Source data from OverPassTurbo, Copernicus and Lithuanian databases.

1- Satellite images :

The table below shows the different satellites used and their advantages and disadvantages for our analysis.

	Landsat 8/9	Sentinel 2A-2B	Kanopus-V
Organisation	NASA/USGS (USA)	ESA (European Union)	Roscosmos (Russia)
Launch Date	L8 : February 2013 L9 : September 2021	S2A : June 2015 S2B : March 2017	July 2012
Spatial Resolution	30 m (Optical MS) 15 m (Pan. Band 8) 100 m (Thermal)	10 m (B2, B3, B4, B8) 20 m (B5, B6, B7, B8A, B11, B12) 60 m (B1, B9, B10)	2.1 m (Panchromatic) 10.5 m (Multispectral)
Revisit	16 days (per satellite) 8 days (L8+L9 combined)	5 days (at equator) 2-3 days at mid-latitudes (2 satellites)	3-5 days (theoretical)
Spectral Bands	11 bands : 4 VIS + 1 NIR (30m) 2 SWIR (30m) 1 Pan (15m) 2 TIR (10.6-12.5 µm)	13 bands : 4 VIS at 10m 6 Red Edge/NIR/SWIR at 20m 3 atmospheric bands at 60m No thermal band	5 bands : 1 Pan (2.1m) 4 MS (Blue, Green, Red, NIR at 10.5m)
Advantages	Thermal bands for urban heat island analysis SWIR bands for urban material discrimination	Better spatial resolution (10m) High revisit frequency "Red Edge" bands for vegetation health	High spatial resolution (2.1m) Good for individual building detection NIR band useful for NDVI
Disadvantages	Coarse resolution (30m) for urban details Low frequency Frequent cloud cover in Baltic region	No thermal band Uneven resolution between bands (10m/20m/60m)	Costly/restricted access Limited spectral bands No SWIR or thermal

By manipulating the spectral bands of each satellite, thanks to the "Image Calculator" function of the TerrSet software, we were able to calculate indices. These are images calculated from satellite images that highlight a specific phenomenon that is present, while attenuating other factors that alter the effects on the image. For example, a vegetation index will present healthy vegetation in a light color in the index image.

Below, we have listed the indices used before presenting them in the form of a table.

- NDVI (Normalized Difference Vegetation Index)
- NDWI (Normalized Difference Water Index)
- MNDWI (Modified Normalized Difference Water Index)
- NDBI (Normalized Difference Built-up Index)
- BSI (Bare Soil Index)

	NDVI	NDBI	NDWI	MNDWI	BSI
Formula	(NIR - Red) / (NIR + Red)	(SWIR1 - NIR) / (SWIR1 + NIR)	(Green - NIR) / (Green + NIR)	(Green - SWIR1) / (Green + SWIR1)	((SWIR1 + Red) - (NIR + Blue)) / ((SWIR1 + Red) + (NIR + Blue))
Purpose	The most classic index for assessing vegetation health, density, and biomass	To map urban and built-up areas	To delineate and monitor open water bodies	An improved water index that better separates water from built-up land by replacing the NIR band with a SWIR band	Designed to enhance the visibility of bare soil and mineral surfaces
Advantages	Easy to calculate and interpret. Values range from –1 to 1 where high value (~0.6 to 0.9) indicate dense, healthy vegetation	Straightforward Calculation : Easy to compute with common satellite bands	Excellent at suppressing vegetation and soil signals, making water stand out clearly	Superior to NDWI : Much more effective at excluding urban and built-up areas, resulting in a cleaner extraction of water bodies	Effective for Soil Mapping: Helps in mapping agricultural fields without cover arid regions, and exposed earth
Disadvantages	In very dense forests or crops, the index can max out and fail to distinguish further growth	Confusion with Bare Soil : Both built-up areas and bare soil have high SWIR	Sensitive to Built-up Areas : Can mistakenly highlight urban surfaces which have a similar spectral response, leading to commission errors	Requires a SWIR band, which is available on most modern satellites (Landsat, Sentinel-2) but not on all	Vulnerable to Confusion : Can be confused with built-up areas (similar spectral response) and is influenced by soil moisture

Finally, we transformed the raw satellite data into usable geographic information through maps, statistics and predictions with the help of the tools for transformation and improvement, classification, forecasting, detection and modeling of changes offered by TerrSet software. Here are the tools used for this analysis :

2- Image Enhancement & Transformation Methods :

Module	Category	Definition	Advantages	Disadvantages
MNF	Preprocessing	Transformation that orders components by signal-to-noise ratio, segregating noise in the data	Better for noisy datasets Preserves meaningful information in first components	More computationally intensive than PCA Still a mathematical abstraction
PCA	Preprocessing	Statistical technique that transforms correlated multispectral bands into uncorrelated components, concentrating variance in the first few components	Reduces data redundancy Useful for change detection Reduces noise	May lose specific spectral information Difficult to interpret components physically
RGB	Visualization	Overlay of 3 bands into the Red, Green, and Blue channels to create a color image	Flexible band combinations for different features Powerful for photo-interpretation and ground truthing	Limited to 3 bands at a time Subjective color interpretation

3- Classification Methods :

Module	Category	Definition	Advantages	Disadvantages
DECISIONFOREST	Supervised Classification	Modified version of the free-standing DecisionForest module. A machine learning method combining many decision trees for a final vote	Very high accuracy, often the top performer Resistant to overfitting Provides variable (band) importance	Can be slow on very large datasets
ISOCLUST	Unsupervised Classification	Unsupervised classification that uses iterative self-organizing clustering to identify natural groupings in spectral data	No training data required Identifie natural spectral classes	Complex parameterization Slower than others Unsupervised Classification Number of classes must be predefined
ISODATA	Unsupervised Classification	Advanced unsupervised clustering that iteratively splits, merges, and eliminates clusters based on statistical parameters	Better for urban objects More sophisticated than ISOCLUSTER Can split and merge classes automatically	More parameters to set Results can vary with parameter choices
KMEANS	Unsupervised Classification	Partitions data into K clusters by minimizing within-cluster distance	Simple algorithm Effective if classes are compact and spherical	Requires predetermined number of clusters Sensitive to initial cluster centers
MLP	Supervised Classification	Supervised neural network classifier or regression with hidden layers that learns non-linear decision boundaries	Good performance on noisy data High classification accuracy	Requires parameter optimization Long training time
SOM	Supervised Classification	Neural network approach that creates ordered mapping of high-dimensional data onto lower-dimensional space	Preserves topological relationships Can reveal subtle patterns No assumption about data distribution	No assumption about data distribution Long training time than DECISIONFOREST

4- Change Detection & Modeling Methods :

Module	Category	Definition	Advantages	Disadvantages
CA_MARKOV	Modeling and Projection	Cellular Automata via Markov Model. Simulates land use/cover change by integrating proximity factors, constraints, and transition potentials.	Integrates spatial drivers (slope, roads, zoning) Produces change probability maps. Good for land-use planning	Requires two historical maps Complex to calibrate. Extrapolates past trends.
IMAGEDIFF	Change Detection	Subtracts pixel values of two images to measure change	Simple and direct. Very effective for abrupt changes Visually intuitive	Highly sensitive to acquisition differences Requires rigorous radiometric correction
LAND CHANGE MODELER (LCM)	Modeling & Projection	Toolkit for analyzing, modeling, and predicting land cover change with multiple algorithms	Integrated workflow Multiple validation tools Combines multiple approaches	Requires quality multi-temporal data Results depend on driver variable selection

Others data

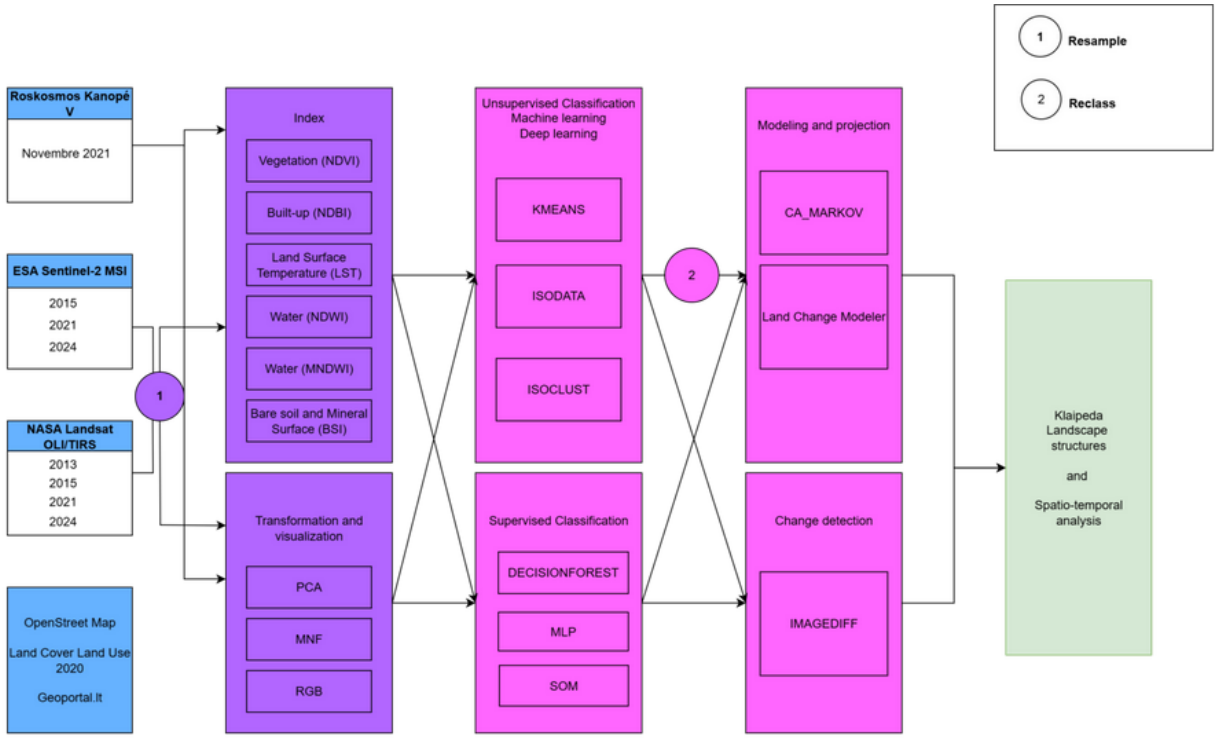
- Open Street Map
- CorineLandCover (in Copernicus)
- Lithuanian databases

Then, we retrieved Open Source data from Open Street Map, CorineLandCover and the Lithuanian database. This allowed us to obtain reference data since it is mainly verified, accurate and recognized official data. They then serve as an example for our analyses and processes carried out on the TerrSet software, allowing us to sometimes observe limits and flaws in certain data. Finally, we were able to use this data to compare it with our results in order to estimate validation percentages on the consistency of the results. The table below presents the advantages and disadvantages of this Openource data for our work.

	Open Street Map	Corine Land Cover	Lithuanian databases
Advantages	Free and open access Frequent updates by local contributors Rich attribute information Contributing characteristics	Free and open access Officially verified and validated data Harmonised nomenclature	Officially verified and validated data High positional accuracy Complete cadastral information Construction Dates and Permits Building heights and surfaces

Disadvantages	No official validation Depends on volunteer effort Often missing construction dates	Coarse spatial resolution Very limited urban details No inter-period information Recent (post-2018) changes not available No distinction between building types	Limited details on small urban structures

Methodological diagram



III - Analysis of results

A. Detection, recognition and identification of urban objects and structures :

In this part, we focused on the urban objects of Klaipeda. First, it was necessary to carry out a typology of urban object types, structures and morphologies for recognising and mapping urban land use types and functions. Finally, we made comparisons between these results.

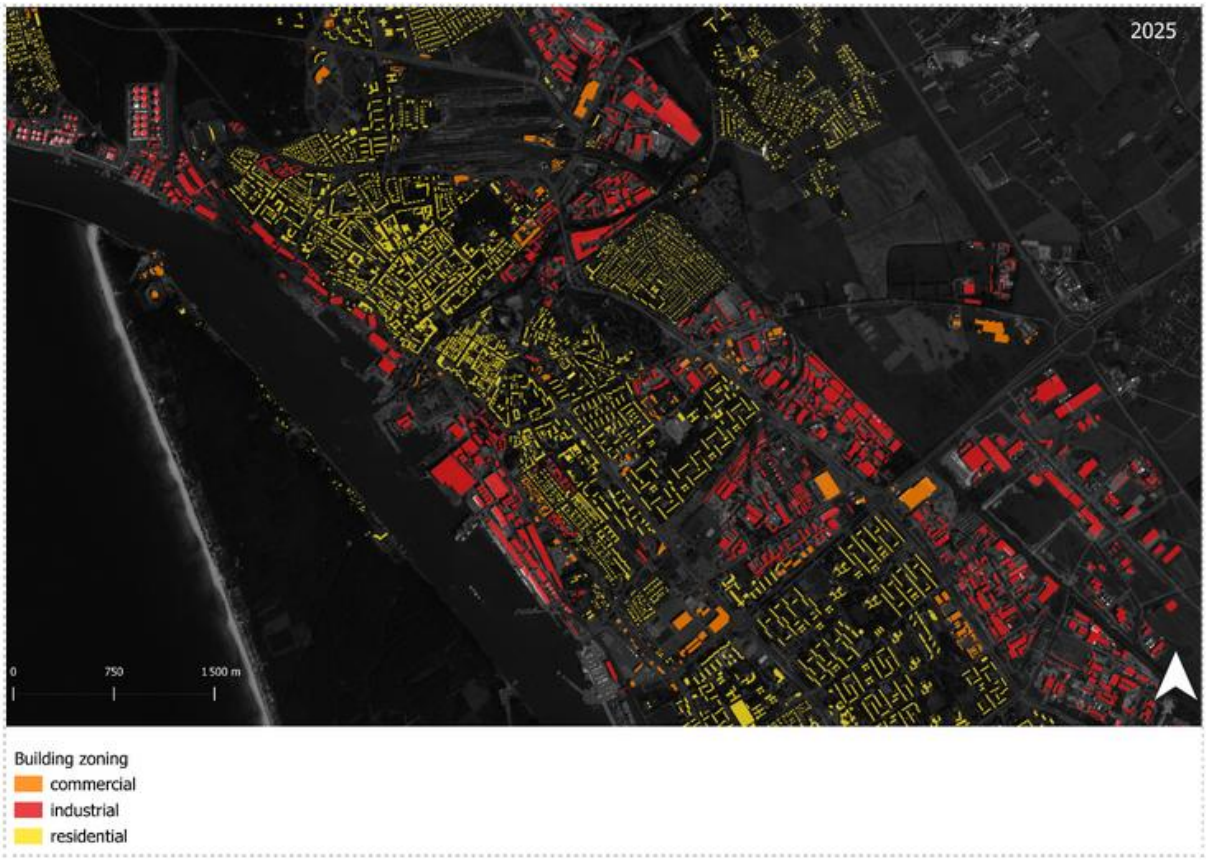


Figure 5 : Urban structure of Klaipėda

The map shows an urban organization typical of a Baltic port city. The visualization reveals a residential building clearly dominating the space, particularly in the heart of the agglomeration. This residential building is formed according to a dense orthogonal architecture, particularly in the centre, probably testifying to the heritage of the city's history. The historic centre is easily identifiable in particular by its morphology characterised by an irregular network of winding and narrow streets, although the city was heavily destroyed during the Second World War, its road network has been reconstructed. It is bounded by the port and the river. To the south-east of the latter, there is a residential area made up of collective complexes, identifiable by the size and atypical shape of the collective buildings. In peri-urban areas, we find spaces made up of individual buildings (in the shape of small dots here on this scale), especially in the north of the city. Industrial areas are mainly polarized on the outskirts to the east, but also along the port, encircling residential areas. It should be noted that this industrial building does not form a single large industrial zone. It is identifiable, for the most part of the buildings, by their size, they are buildings with a large footprint designed to house storage lines, large machines or production, but also by the shape and material of their roofs.

Commercial buildings, which are more dispersed, are mostly located along the main roads, often near residential buildings, which suggests that access to local services is easier, thus reducing travel for services and everyday purchases. We therefore have a development policy that favours contact between place of residence, work and consumption while maintaining a certain isolation from industrial activities.

However, this proximity between the residential and industrial sectors can raise problems of noise, traffic or atmospheric pollution, or otherwise problems of incompatible uses. The city must therefore pursue development policies that reduce these problems of use, but it must also find a way to reconcile its port economic activities with its urban quality of life in order to ensure its preservation. This could potentially push Klaipeda to face the challenge of the ecological transition, which will involve decarbonisation objectives.

B. Temporal analysis of recent changes in Klaipėda :

In this part, we have Create a time series of Sentinel 2 MSI and Landsat 8/9 OLI/TIRS data from 2013 to 2025 at the same period to observe the evolution of the territory, identify the types of changes. Finally, we made comparisons between these results.

1- Landsat 8/9 OLI/TIRS

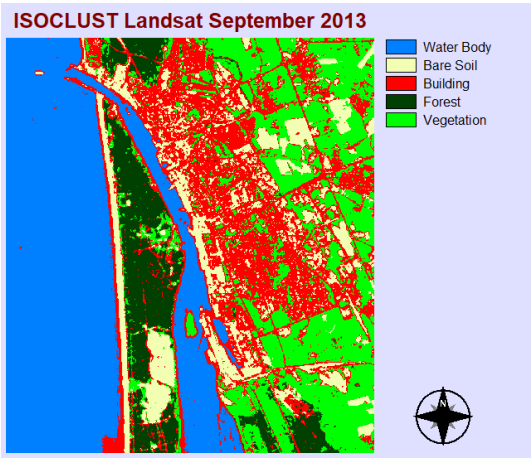


Figure 5 : Vue satellite de Klaipéda (2013)

We have chosen to keep 4 dates : 2013 ; 2015 ; 2021 ; 2024. These images were taken during the same period (between September and November). After having carried out several classification methods for each date, we made projections for the future but also projections for 2024-2025 and 2035-2055 in order to observe (or not) similarities between the predictions and reality.

- **2013**

a- Unsupervised Classification



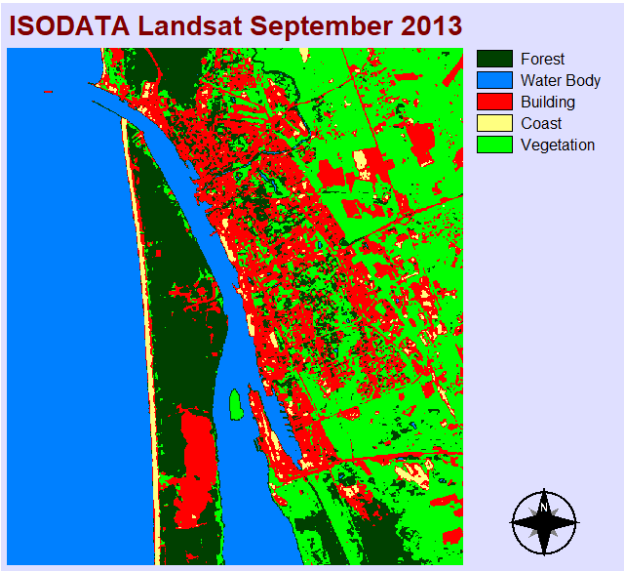
This ISOCLUST was made from 7 spectral bands of Landsat (B2-B8), 70 iterations and 5 desired output clusters.

The result shows more or less homogeneous classes with little noise and fairly well differentiated.

Area on file: ISOCLUST 5 classes

Category	Hectares	Legend
1	4873,027500	Water Body
2	1794,105000	Bare Soil
3	3571,807500	Building
4	1327,567500	Forest
5	3158,842500	Vegetation

The Water Body does not differentiate between the Baltic Sea, the Curonian Lagoon and a few inland watering holes. The boundary between land and sea is clear and precise. Bare Soil has a rather fragmented distribution with possibly agricultural areas without vegetation, land under construction, or sandy areas. Some places could be mistaken for light roofs of buildings or industrial and port areas. The Building occupies an important place at ISOCLUST. It mainly contains residential, industrial and port areas. However, some soils were considered Building, including the road and part of the sandy coast. The Forest is well detected with a logical spatial distribution and a clear boundary with urban areas. Finally, the Vegetation has a consistent distribution taking into account cultivated areas, meadows and urban green spaces. The distinction between the Forest and the Vegetation is a good one.



This ISODATA was made from 7 spectral bands of Landsat (B2-B8), 70 iterations and an initial and output number of 5 clusters.

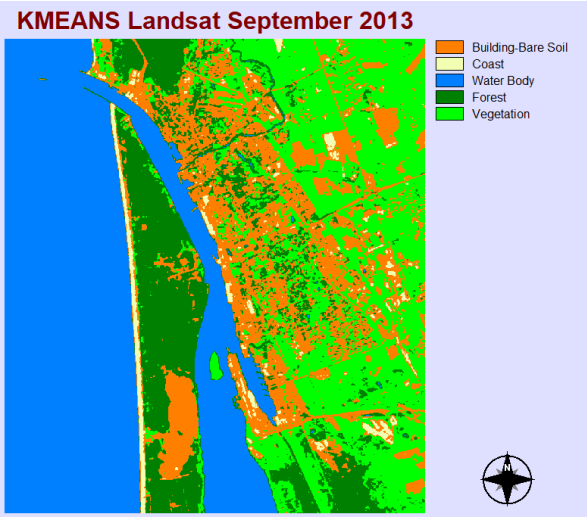
The result demonstrates an acceptable overall spatial coherence with more or less homogeneous

Area on file: ISODATA 5 classes

Category	Hectares	Legend
1	2362,905000	
2	5304,622500	
3	2887,627500	
4	308,385000	
5	3861,810000	

The Forest is well identified, as is the ISOCLUST.

The spatial continuity is remarkable. It should also be noted that there are small urban woodlands. The Water Body is also similar to the ISOCLUST, except for the harbour which has been well identified, as well as the canal to the north of the town and some inland waterholes. As for the Building, it has significant confusion with the Bare Soil and part of the Vegetation but still identified urban and industrial areas. The Coast, a new cluster, represents the coastal sandy area, as well as white roofs and some of the industrial buildings. Finally, the Vegetation has a good identification despite nuances. This cluster is similar to that of ISOCLUST, however, some parts of the Bare Soil have been confused with the Vegetation.



This KMEANS was made from 7 spectral bands of Landsat (B2-B8), 70 iterations and an initial and output number of 5 clusters. The result demonstrates a more spatially homogeneous classification.

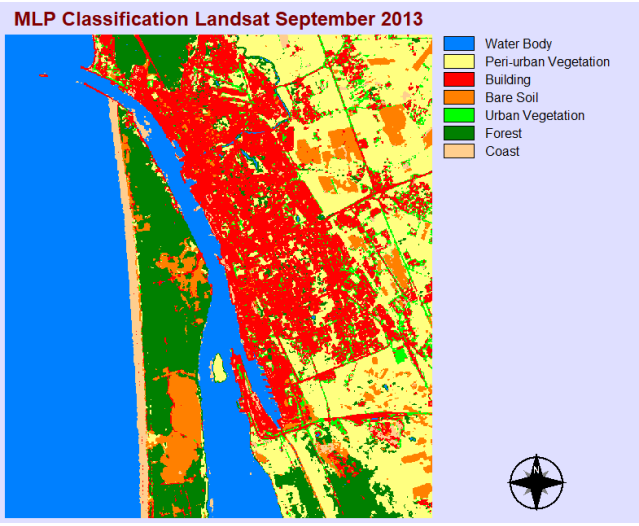
Area on file: KMEANS 5 classes

Category	Hectares
1	2889,945000
2	301,657500
3	5304,442500
4	2365,807500
5	3863,497500

The Building and the Bare Soil have been grouped together, as in ISODATA above. The distinction was once again not made. The Coast has a similarity to ISODATA with the same qualities and flaws. This goes hand in hand with the Water Body, the Forest and the Vegetation.

These 3 unsupervised classifications show us more or less different results. The ISOCLUST is the only algorithm that distinguishes between Building and Bare Soil. However, ISODATA and KMEANS manage to distinguish between Bare Soil and the Coast. In addition, despite the distinction made by ISOCLUST, it can be observed that the Building cluster has an area just as large as the surface area of Building and Bare Soil combined in ISODATA and KMEANS. Thus, these 3 algorithms propose their own interpretation, marked by the spectral variability of geographical objects where different objects can have similar spectral signatures. Although all three methods use the same input parameters, they produce entirely different results, thus demonstrating the internal logic of the algorithm.

b- Supervised Classification



This MLP was made from 7 spectral bands of Landsat (B2-B8) and 5 indices (MNDWI, NDBI, BSI, NDVI, NDWI).

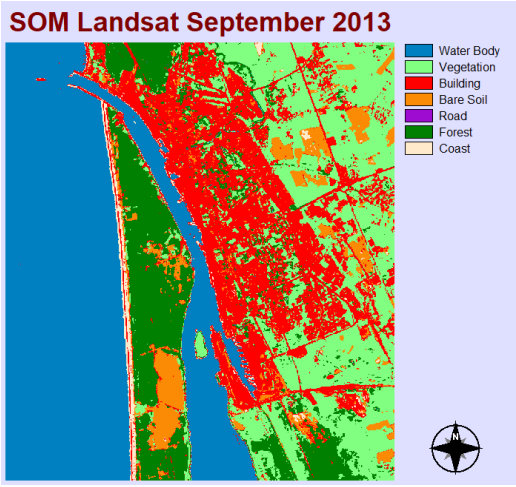
The result demonstrates, through a sophisticated and multivariate approach, a methodological improvement compared to previous unsupervised classifications

Area on file: MLP Classification Bandes Spectrales et Indices

Category	Hectares	Legend
1	5186,835000	
2	2961,180000	
3	3031,762500	
4	880,695000	
5	516,172500	
6	1698,817500	
7	449,887500	

Thanks to the MNDWI and NDWI indices, the MLP has made a distinction between the lagoon and the sea for the Water Body cluster. The boundary between land and water is extremely clear. The NDVI made it possible to differentiate between the Peri-urban Vegetation and the Urban Vegetation with large plots facing small plots, but also less dense vegetation facing very dense vegetation. In addition, by introducing NDBI, the algorithm managed to identify the Building rather adequately. Industrial and port areas, the city centre and residential areas are always taken into account. This time, we can also

note the presence of recent developments in the built environment with the characterization of some peri-urban extensions. Therefore, the Bare Soil is differentiated from the Building thanks to the BSI. However, it should be nuanced with the vegetation that sometimes takes over. Concerning the Forest, the spatial distribution is still as coherent as ever, always taking into account some urban woodlands. Finally, the Coast has been detected along the Baltic Sea.



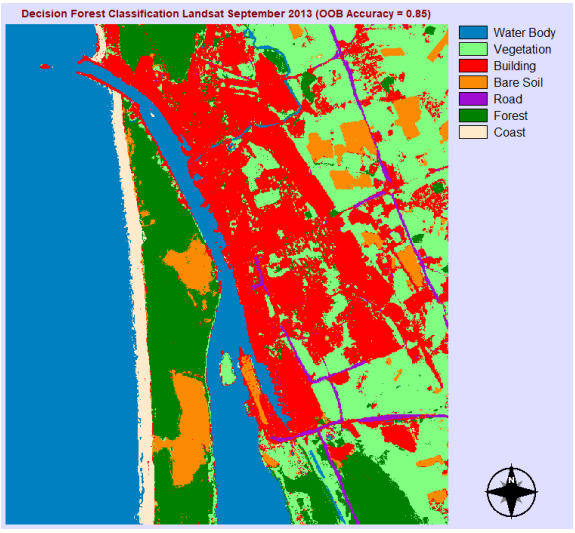
This SOM was made from 7 spectral bands of Landsat (B2-B8), 5 indices (MNDWI, NDBI, BSI, NDVI, NDWI) and a MAKESIG with 7 clusters.

The result is an overall satisfactory classification with a good performance for the natural classes (Water Body, Vegetation and Forest. On the other hand, the performance is not as good for the rest.

Area on file: SOM Bandes Spectrales et Indices

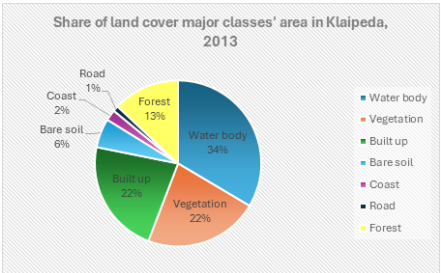
Category	Hectares	Legend
1	5152,095000	
2	3354,007500	
3	3241,890000	
4	839,902500	
6	1921,612500	
7	215,842500	

The SOM perfectly identifies the Water Body with the Baltic Sea, the central channel and the water bodies, thanks in particular to the MNDWI and NDVI indices. With the NDVI, the Vegetation is well recognized. For the Building, the general urban structure is well captured, the NDBI index worked quite well. However, this cluster indicates an over-classification, confusion is made between certain bare grounds, car parks and Road, a cluster that has not been identified by the SOM. The Forest has been detected. The dune forest strip to the west is well identified, there is a good separation with the herbaceous/agricultural vegetation. Finally, the coastal beach is identified in beige, corresponding to the light sandy areas. The land-sea interface is well defined.



This DECISIONFOREST was made from 7 spectral bands of Landsat (B2-B8), 5 indices (MNDWI, NDBI, BSI, NDVI, NDWI) and a MAKESIG with 7 clusters.

The classification by Decision Forest has an OOB (Out-Of-Bag) accuracy of 0.85 (85%), which is an excellent performance indicator

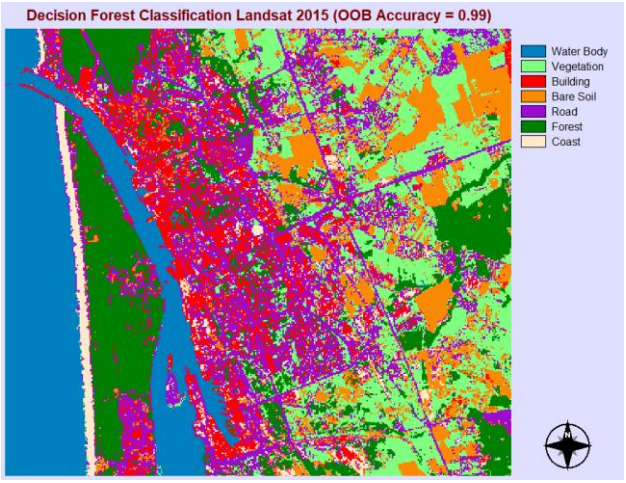


The Water Body has a better classification than the SOM since we can distinguish 2 channels located in the North and South. The Vegetation mixes with the Building, especially in the city centre. The Bare Soil is less fragmented. The Road has been detected where major roads can be distinguished. The Forest is less pronounced here in the city center. It is still confused with the Vegetation. Finally, the Coast is better represented, the distinction with the Building is well made.

These 3 supervised classifications show us more or less different results. They create a complete processing chain. Only the DECISIONFOREST has succeeded in distinguishing the Road, in particular due to the fact that the spatial context is very little taken into account. In SOM, semantic flexibility is complex. As for the MLP, it differentiates the type of vegetation

- **2015**

a- Supervised Classification



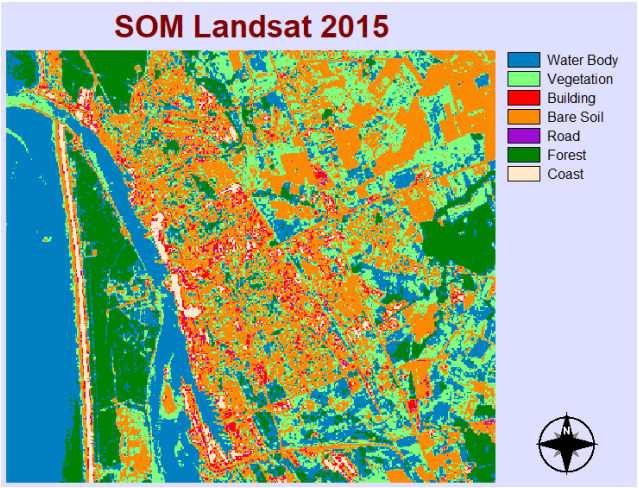
This DECISIONFOREST was made from 3 spectral bands of Landsat (B2, B3 et B4) and a MAKESIG with 7 clusters.

The result is not very suitable, without any spatial coherence.

Area on file: FOREST__df

Category	Hectares
1	2583,607500
2	2637,697500
3	1503,945000
4	1771,425000
5	2909,700000
6	2353,230000
7	531,315000

In this classification, the Road has been very poorly identified. This is the cluster with the largest area, which makes us note that the result is bad, despite the recognition of the Water Body, the Vegetation, the Bare Soil and the Forest. The Building is also subject to poor spatial coherence. Urban areas are poorly distinguished and peri-urban areas are not at all. Finally, the Coast was distinguished, but some areas are also part of this classification, notably in the port and the industrial zone. Therefore, this result is bad.



This SOM was made from 3 spectral bands of Landsat (B2, B3 et B4) and a MAKESIG with 7 clusters.

The result is worse than the DECISIONFOREST, with spatially inconsistent classifications.

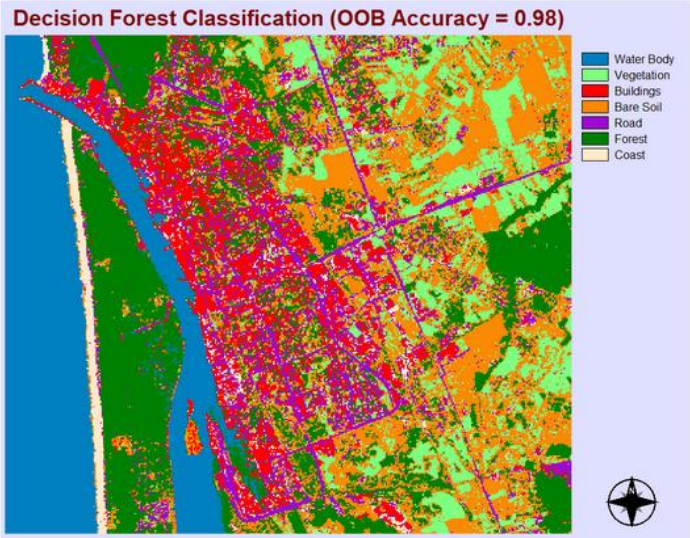
Area on file: SOM_2015_RGB_Supervised

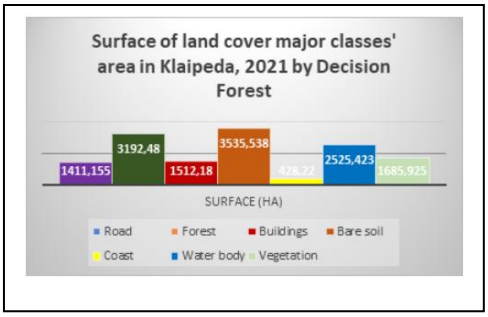
Category	Hectares	Legend
1	4224,217500	
2	2907,877500	
3	385,830000	
4	4359,240000	
5	155,475000	
6	1821,465000	
7	436,815000	

In this SOM, only the Forest is more or less well represented. The rest is misclassified. The Road has not even been identified. The result is once again bad.

- **2021**

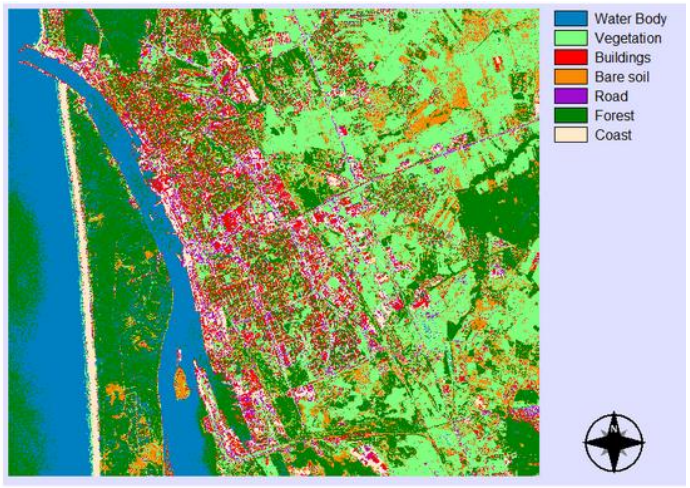
a) Supervised Classification

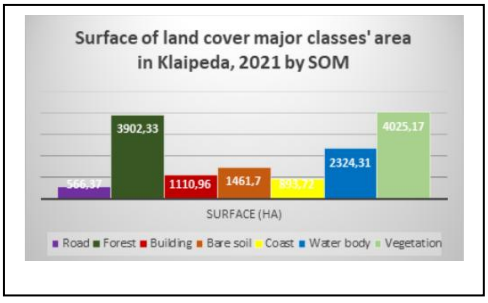




Once again, the "Decision Forest" method gives very good results, with excellent performance, illustrating the remarkable reliability of the Machine Learning algorithm. These results highlight the occupation of the territory of Klaipėda, a Baltic port city, with a clear distinction between the major spatial areas of occupation: the body of water in the west, the urban fabric at the heart of the map and the green spaces that dominate the peri-urban and forest areas. Once again, bare soils probably reveal agricultural areas or simply areas in urban transition, identifiable in particular thanks to infrared data from Landsat satellites.

However, although the OOB accuracy is remarkable, with a value of 0.98, the visualization shows a "salt and pepper" effect, especially in anthropogenic spaces where the pixels of the different sets intermingle in a fragmented way. This is most likely due to the low resolution of the Landsat satellite bands. At this resolution, a pixel can contain different elements of spatial occupation (vegetation, roof, road, etc.), thus generating mixed spectral signatures that reveal this fragmentation in heterogeneous environments, limiting the fine interpretation of the spatial structure of the territory, particularly urbanized spaces. But this method remains, despite everything, interesting for the analysis of global spatial occupation, as can be seen with forest areas, which are remarkably accurate.





The classification by Self-Organizing Maps applied to the RGB result of the Landsat bands offers broadly the same results as the previous method. Indeed, the visualization clearly makes it possible to identify and distinguish the plans present in the region, in particular that of the Curonian lagoon, the vegetation, the urban spaces, marked by a significant presence of buildings in industrial areas, the city

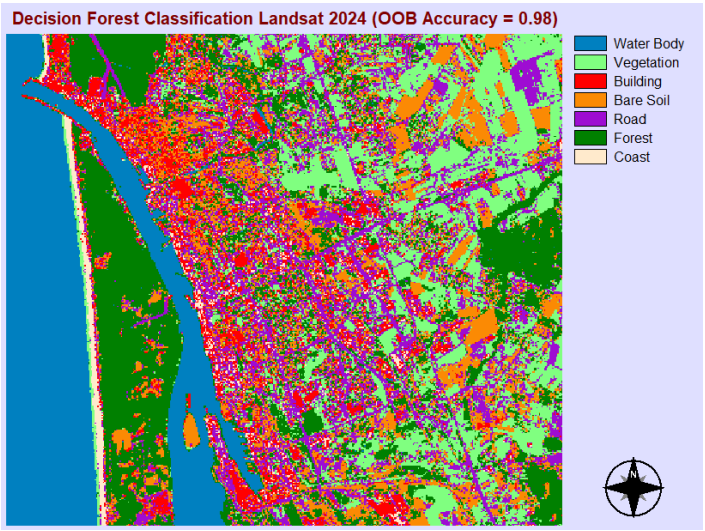
center but also residential areas. But above all, it makes it possible to identify forest areas. The distinction between low vegetation and forest is very interesting because it makes it possible to differentiate vegetated areas according to their nature or density. The road network, although very fragmented, is easily identifiable, as well as the bare ground areas, mainly occupying the eastern and southern zones.

Nevertheless, despite the fact that this classification is more or less valid, we find, here too, the "salt and pepper" effect, which remains very pronounced, more precisely in dense urban spaces. The result is a significant fragmentation of classes, giving rise to spectral confusions, particularly between roads and buildings, but also between bare soils, the "coast" and vegetation in urban spaces.

Although it is interesting to provide the natural structure of spectral data, the SOM method does not offer a robust enough quantitative validation like the OOB precision in Decision Forest. As can be seen on the map, this classification by Self-Organizing Maps reveals several errors related to the spectral signatures of certain elements, in particular the presence of elements of the "Forest" class in an area in the Baltic Sea to the west of the city, but also the confusion between the built environment and the "Coast".

In any case, regardless of the method used, post-processing of the data is still necessary to reduce the fragmentation related to the resolution of the Landsat OLI/TIRS satellite bands, thus offering much more usable graphical results.

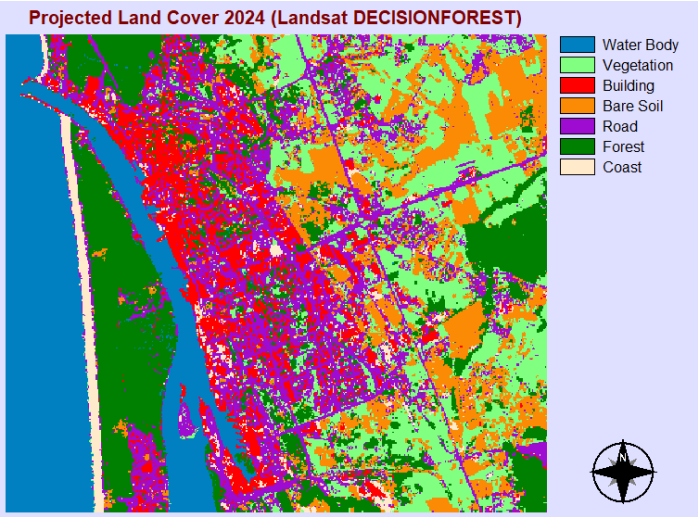
b- Modeling and Projection



Area on file: Forest_Landsat_2024_df

Category	Hectares
1	2581,582500
2	2638,192500
3	1198,395000
4	2049,345000
5	2536,605000
6	2965,815000
7	320,985000

This is a DECISIONFOREST made from 3 spectral bands of Landsat (B2, B3 and B4) in 2024. We then compared this image to the projection CA_MARKOV made using the 2015 and 2021 DECISIONFOREST to find out whether or not the result of the CA_MARKOV is consistent with the 2024 DECISIONFOREST.



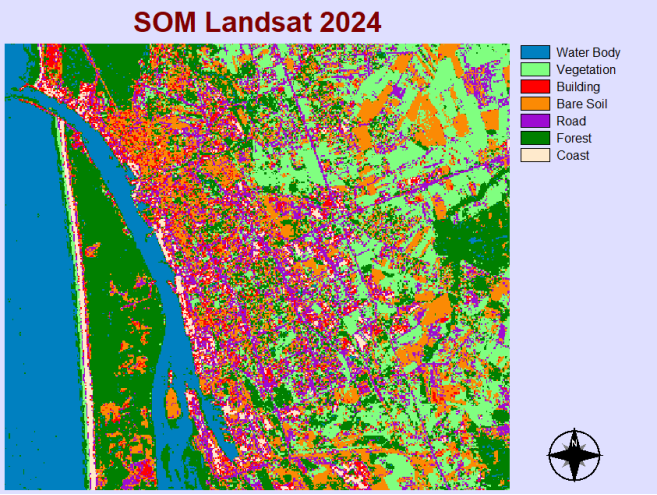
Area on file:
CA_Markov_L8_2015_2021_Prevision3ans5x5

Category	Hectares
1	2553,007500
2	2519,122500
3	1417,545000
4	2003,625000
5	2680,425000
6	2631,105000
7	486,090000

Comparing the acreage results, the prediction is roughly good for the Water Body, the Vegetation, the Bare Soil, and the Road. The results are more or less equivalent. On the other hand, the Building is over-represented, unlike the Forest which is under-represented. The Coast cluster has undergone an increase in prediction that should not be overlooked. All in all, the prediction is quite correct, even if some clusters have a different representation. This prediction still needs to be qualified for clusters with equivalent results since in some places, the land cover is different compared to the real DECISIONFOREST, especially for the Vegetation and the Bare Soil.

- **2024**

a- Supervised Classification



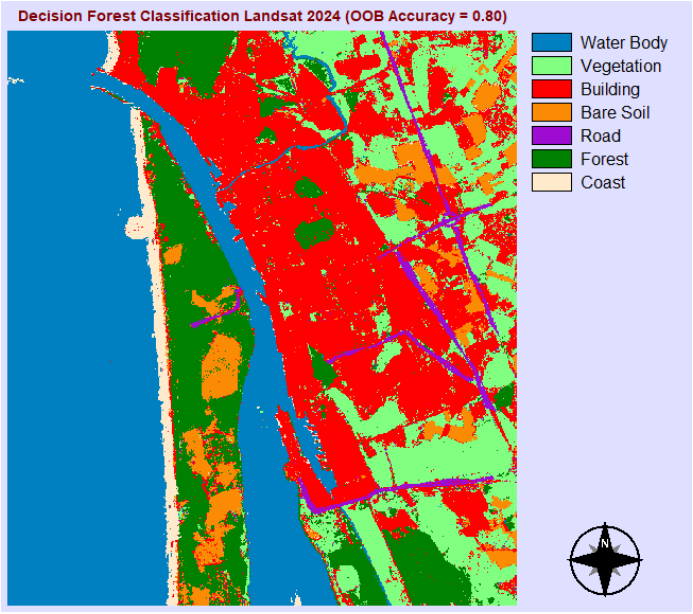
This SOM was made from 3 spectral bands of Landsat (B2, B3 et B4) and a MAKESIG with 7 clusters.

Le résultat est semi-réaliste, avec des classes plus ou moins plausibles.

Area on file:
SOM_RGB_Landsat_2024_Supervised

Category	Hectares	Legend
1	2431,867500	
2	2883,172500	
3	838,125000	
4	2524,050000	
5	1902,645000	
6	3298,950000	
7	412,110000	

The Water Body has been captured, whether it is the Sea, the lagoon or the land waterholes. The vegetation is moderately realized. Peri-urban vegetation is well characterized, unlike urban vegetation which is not. The Building is not well represented and has been partly confused with the Bare Soil, hence the large surface area of this cluster. The Roas is also overestimated, if you can make out the road network, you can see plots in the city centre and on the outskirts. The Forest is well detected, as for the Coast, the sandy area is represented, but the industrial and port areas are also represented, as are some buildings.



This DECISIONFOREST was made from 7 spectral bands of Landsat (B2-B8), 5 indices (MNDWI, NDBI, BSI, NDVI, NDWI) and a MAKESIG with 7 clusters.

Le résultat présente une bonne robustesse, comme l'indique l'OBB. Les classes sont plus distinctes.

Area on file: DecisionForest Indices_df

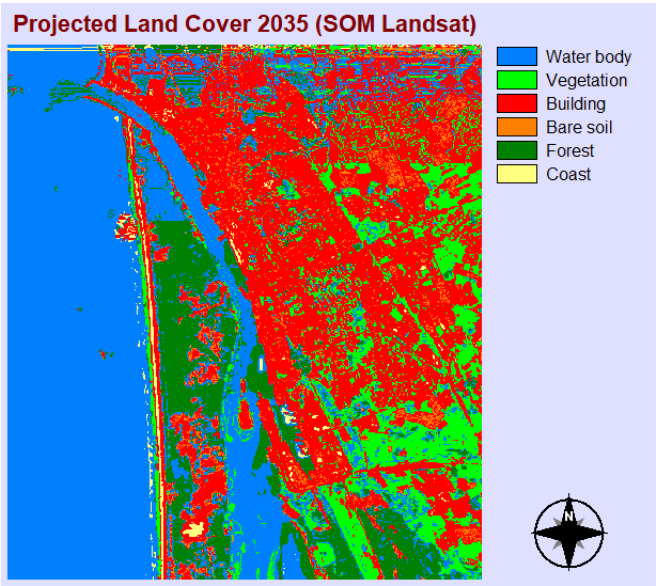
Category	Hectares
1	5195,745000
2	2201,062500
3	4020,952500
4	824,962500
5	278,662500
6	1875,240000
7	328,725000

The Water Body is still well represented. The Vegetation, with the help of NDVI, is well distinguished from the Bare Soil. There are small areas of the Vegetation in the city center, like for the Forest. The Building has a good spatial coherence both in the centre and on the outskirts. As for the Road, it is better represented. Finally, the Coast is only located on sandy areas, showing the good realism of this cluster.

b- Modeling and Projection

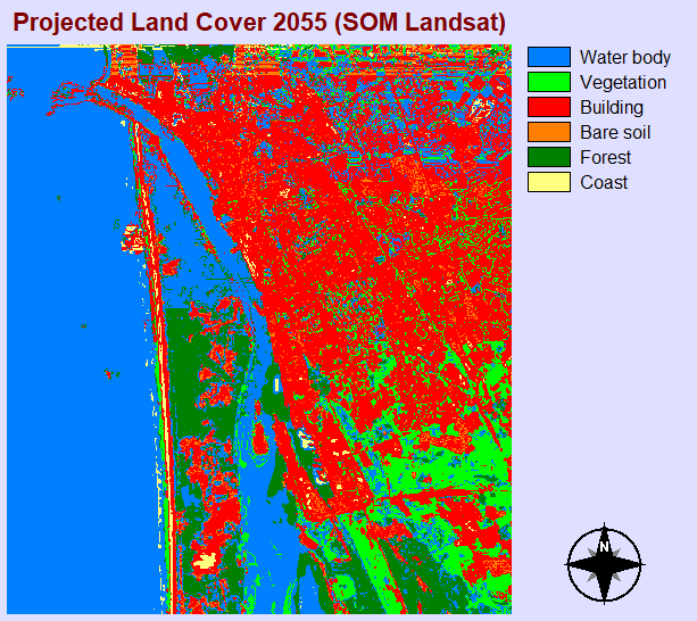
From now on, we wanted to make long-term predictions, through the CA_MARKOV tool, using the DECISIONFOREST of 2013 and 2024 in order to estimate land use for the years 2035 and 2055, in contexts marked by urban sprawl and sea level rise, 2 phenomena that can impact Klaipéda and its territory.

The Markov model application shows a nuanced projection of how land use changes over thirty years, highlighting continuity and transition dynamics. The classifications established in 2013 and 2024 served as the basis for this projection and reveal a landscape with a strong presence of dense natural masses. The results indicate an overall persistence of this fundamental structure, dominated by forests, vegetation cover and water bodies, thus highlighting the resilience of these large complexes to the pressure of urbanization in the short and medium term.



The 2035 projection presents a first stage in the evolution compared to 2025. It can be seen that the fundamental structure of the region, which are the forest areas, the water bodies and the vegetation cover, remains generally preserved, thus highlighting a certain inertia of these major ensembles. However, the same cannot be said for bare surfaces. They indicate major spatial variations, which may reflect the dynamics of construction sites, temporary abandonment or land clearing. The latter seem to

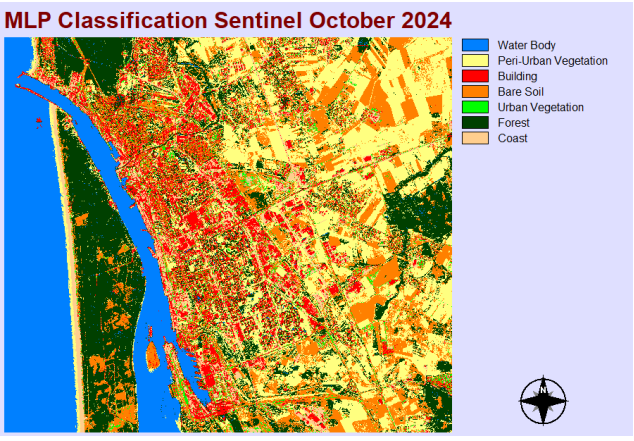
experience localized gains, with probable extensions and/or densifications at the periphery of existing nuclei, which follow the calculated transition probabilities. This projection marks a period of moderate change, in which the trends of territorial change, despite the fact that they have already begun, have not yet dramatically altered the balance of the territory. It represents a crucial step, which validates the orientation of the model before the longer-term projection.



The projection for 2055 shows an exacerbation and a reinforcement of the dynamics started in the 2035 projection, giving a vision of a region whose land use has undergone significant changes over three decades. The resilience of large natural areas, wetlands, forests and coastlines, consolidates their sustainable disposition, despite the fact that there is localized retreat and fragmentation in their fine distribution, particularly in areas close to areas of urban pressure. Concerning artificialized areas, we observe a more marked, densified and more saturated morphology, particularly towards the east of the territory. Indeed, the urban task tends to expand towards unbuilt spaces according to the logic of spatial continuity, thus often reducing bare ground spaces. We can also note the decline of low vegetation in favour of the extension of buildings. This projection shows a clear shift in the balance between natural and artificial in some areas of the Klaipeda region, which materializes the cumulative influence of thirty years of transitions and modelled changes, and underlines the influence of the trends present in the projection and future organization of this region.

2- Sentinel 2 MSI

a- Supervised Classification



Made with Sentinel B2, B3 and B4 tapes.

The map is consistent for a classification with 3 visible bands, but some classes seem to be not very distinct spectrally

Area on file: MLP_2024LULC_Test_812

Category	Hectares	Legend
1	2559,440000	
2	3596,230000	
3	1470,280000	
4	2529,330000	
5	418,430000	
6	2786,400000	
7	924,450000	

The Water Body has a good realism taking into account the sea and the lagoon. A distinction has been made between the Vegetation, however, the Urban Vegetation is very much in the minority. The

representation of the Bare Soil is correct. Concerning The Forest, the precision is good, the same for The Coast.

Conclusion

This comparative analysis of the performance of the Landsat OLI/TIRS (NASA), Sentinel-2 MSI (ESA) and Kanopus-V (Roskosmos) satellites over the city of Klaipeda has made it possible to evaluate the differentiated capabilities of these three observation systems in two complementary areas: the fine detection of urban structures and the temporal monitoring of land cover dynamics. The complementarity between these three systems becomes particularly evident in a multi-scale and multi-temporal approach to the urban analysis of the territory. In addition, this work required the methodological integration of multi-source data. The differences in spatial, spectral, and radiometric resolution between the three sensors require rigorous harmonization protocols to allow for comparative or merged analyses. Land cover classification methods, whether traditional supervised approaches, neural networks or self-organizing maps, must be adapted to the specific characteristics of each sensor, all in a city where the complexity of the urban fabric mixing residential areas with varying densities, port and industrial facilities, urban green spaces, forest corridors and peri-urban agricultural areas, is a special case study.

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